

Extending RotatE: Interpretability and Multi-Relational Modeling

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Abstract—Knowledge graph embedding models have shown great promise in representing entities and relations for downstream reasoning tasks. Among these, RotatE effectively models relation patterns as rotations in the complex plane but suffers from limitations in handling *non-commutative relations*, *multi-fold relations*, and *imbalanced relation distributions*. In this work, we conduct a thorough study of recent extensions, and introduce a series of targeted enhancements to address these challenges. Specifically, we integrate *dual-geometry entity rotation* to better model one-to-many, many-to-one, and many-to-many relations, adopt *relation-balanced sampling* to mitigate the impact of relation frequency imbalance during training, and implement *adversarial temperature annealing* to stabilize negative sampling and accelerate convergence. Our careful implementation,¹ analysis, and combination of these techniques allow the model to capture complex relational patterns with higher fidelity and robustness. This work demonstrates that systematic enhancements to the RotatE framework can substantially improve performance on complex and imbalanced knowledge graphs, providing a strong foundation for further exploration in high-precision knowledge representation.

I. INTRODUCTION

Knowledge graphs (KGs) have emerged as indispensable tools for representing structured relational data across diverse domains, ranging from natural language processing and recommendation systems to biomedical and legal reasoning. Embedding models for KGs aim to project entities and relations into continuous vector spaces, enabling efficient reasoning and link prediction. Among these, the RotatE model [1] has gained substantial attention for its elegant geometric formulation, modeling relations as rotations in the complex plane and effectively capturing relational patterns such as symmetry, inversion, and composition.

Although many recent approaches achieve remarkable predictive performance, **interpretability continues to be a vital yet frequently overlooked challenge**. Comprehending *why* a model predicts a specific relation is especially important in high-stakes domains such as healthcare, finance, law, and scientific research, where transparent reasoning can inform critical decisions and reveal hidden patterns within relational data. Recent advancements using graph convolutional networks (GCNs), such as R-GCN [5] and CompGCN [4],

have demonstrated strong results by incorporating relation-specific transformations and composition-based message passing. However, these methods often sacrifice transparency: the deeper the network, the less interpretable the reasoning behind individual predictions.

To address these challenges, our work builds upon two significant extensions of RotatE: RotatE-CT [12] and MRotatE [13]. RotatE-CT introduces a *translation component* alongside rotation to handle non-commutative relations, allowing the model to differentiate cases where relation order matters. MRotatE further incorporates an *entity rotation mechanism* to address multi-fold relations, enabling accurate modeling of one-to-many, many-to-one, and many-to-many interactions. Our approach carefully integrates these extensions while introducing *relation-balanced sampling* and annealing of adversarial temperature, mitigating overfitting to frequent relations and improving training stability.

By preserving explicit geometric structures in the embedding space, our method ensures **interpretability** across complex relational patterns. This emphasis on transparent and analyzable representations distinguishes our work from deep GCN-based models, providing insights into both the *mechanics* of KG embeddings and the relational semantics they capture.

II. RELATED WORK

Knowledge Graph Embedding (KGE) has witnessed substantial progress over the past decade, with models aiming to represent entities and relations in continuous vector spaces for efficient reasoning. Early approaches such as TransE [10] modeled relations as translations in the embedding space, capturing simple relational patterns but struggling with complex relation types like one-to-many or many-to-many. DistMult [11] and ComplEx [7] extended this idea by using bilinear or complex-valued interactions, allowing better modeling of asymmetric relations, yet they remain limited in handling non-commutative relation patterns.

RotatE [1] introduced a significant advancement by representing relations as rotations in the complex space, which enabled the modeling of various relation patterns, including symmetry, antisymmetry, inversion, and composition. While RotatE demonstrated strong performance, it struggles with *non-commutative relations* and multi-fold relational patterns, where a simple rotation is insufficient to capture the diversity of entity interactions.

¹Code available at <https://github.com/aminhalvaei/SEN-Project>

To address the limitation in modeling compositional and order-sensitive relations, RotatE-CT [3] combines rotation with translation, effectively introducing a *translation bias* that allows the model to distinguish between relations like “mother of spouse” and “spouse of mother”. This modification helps capture non-commutative relational patterns more accurately, albeit at the cost of added computational complexity.

MRotatE [2] further extends the RotatE framework by introducing *entity rotation*, allowing head and tail entities to rotate under distance constraints relative to their relations. This mechanism enables the modeling of one-to-many, many-to-one, and many-to-many relations more effectively than RotatE or RotatE-CT. Additionally, MRotatE employs *relation-balanced sampling*, which mitigates the effect of relation frequency imbalance during training, improving performance on rare relations.

Rotate3D [9] generalizes rotational embeddings into three-dimensional space, allowing richer geometric transformations to model complex relational patterns and compositional structures. This 3D rotation enables the model to better capture multi-fold relations and higher-order dependencies, though it increases computational complexity and memory requirements compared to 2D rotations.

Tensor-based methods, such as Tucker [6], represent entities and relations using a core tensor and factor matrices, allowing full interaction modeling across all dimensions. Tucker can capture complex relational patterns and multi-fold relations effectively but introduces substantial memory and computational overhead, particularly on large knowledge graphs.

Quaternion-based KGE models, such as QuatE [8] and QuatD [15], extend the idea of complex-valued embeddings to four-dimensional hyper-complex numbers, allowing more flexible rotation operations to capture sophisticated relational patterns. However, the introduction of quaternions also increases both the computational and memory complexity of the models.

Other recent approaches include Graph Convolutional Networks (GCNs) and their variants applied to knowledge graphs, such as R-GCN [14], which aggregate information from neighboring nodes to capture multi-hop relational patterns. Although GCN-based methods often achieve strong predictive performance, they typically sacrifice interpretability, making it challenging to explain specific relation predictions. This limitation motivates our focus on enhancing both the expressiveness and interpretability of rotational embedding methods.

In summary, while many prior works improve the predictive capability of KGE models, challenges remain in capturing complex relational structures, handling multi-fold relations, and maintaining interpretability. Our work builds on ComplEx, RotatE, RotatE-CT, MRotatE, and Rotate3D, aiming to combine their strengths while addressing their respective limitations.

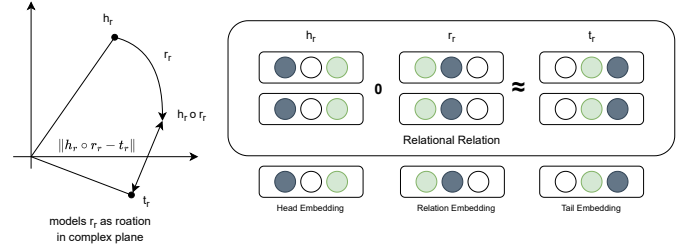


Fig. 1. Relational rotation

III. METHODS

Our work builds on RotatE, a knowledge graph embedding (KGE) model that represents relations as rotations in complex space. While RotatE captures fundamental relation patterns (symmetry, inversion, and simple compositions), it struggles with **non-commutative relations**, **multi-fold relations** (1-to-N, N-to-1, N-to-N), and **imbalanced relation frequencies** during training. To address these challenges, we studied and partially implemented two recent extensions—**RotatE-CT** and **MRotatE**—and introduced two novel modifications: **Relation-Balanced Sampling** and **Adversarial Temperature Annealing**.

A. RotatE-CT: Capturing Non-Commutative Relations

RotatE models all relations as commutative rotations in the complex plane, making it difficult to distinguish between relation orderings (e.g., “mother of spouse” vs. “spouse of mother”). RotatE-CT addresses this by augmenting relation embeddings with a **translation bias**:

$$t = h \circ r + b_r \quad (1)$$

Here, $h, r, t \in \mathbb{C}^d$ are head, relation, and tail embeddings, and $b_r \in \mathbb{C}^d$ introduces translation. By combining rotation with translation, RotatE-CT breaks commutativity, allowing the model to differentiate between order-sensitive compositions.

B. MRotatE: Dual Rotation for Multi-Fold Relations

To better capture multi-fold relations, MRotatE introduces an **entity rotation mechanism** in addition to RotatE’s relational rotation. Entities and relations are mapped into Euclidean space, and for a triple (h, r, t) , the model enforces a distance constraint:

$$\|h_e - t_e\| = \|r_e\| \quad (2)$$

This allows entities to rotate around each other on a hypersphere of radius $\|r_e\|$, rather than being mapped to a single deterministic point. The entity rotation score is defined as:

$$f_e(h_e, t_e) = \|h_e - t_e\| - \|r_e\| \quad (3)$$

By combining this entity-based score with the original relational rotation score, MRotatE creates a **dual-geometry**

embedding space, which improves the modeling of 1-to-N, N-to-1, and N-to-N relations.

Ablation Study

To understand the contribution of each proposed component, we conducted an ablation study on the FB15k-237 dataset at step 4999. We evaluated four settings: the baseline RotatE model, RotatE with the Coordinate Transformation (CT) module, MRotatE, and a combined model integrating both modules.

Table I summarizes the results. Adding the CT module significantly reduces the Mean Rank (MR) from 1010.86 to 325.95, indicating more accurate entity ranking. The MRotatE extension further improves MRR and HITS@k scores, showing its effectiveness in capturing richer relational patterns. When combining both enhancements, we achieve the best performance, with MRR increasing by over 40% relative to the baseline and MR dropping by more than 75%. This demonstrates that the two modules complement each other, leading to stronger link prediction performance.

C. Relation-Balanced Sampling (Explored, Not Adopted)

Knowledge graphs are highly imbalanced, with frequent relations dominating training. Standard RotatE sampling uniformly selects triples, which biases learning towards common relations. We proposed **Relation-Balanced Sampling**, which:

- 1) Samples a relation uniformly from the set of all relations.
- 2) Samples a triple uniformly from triples associated with that relation.

This ensures that each relation contributes more equally to training, improving representation quality for low-frequency relations. We implemented this in a custom `RelationBalancedSampler` class, replacing the default random shuffle step in training.

We experimented with a custom *Relation-Balanced Sampling* strategy designed to reduce bias towards frequent relations by ensuring uniform sampling across relations. However, as shown in Table II, this method performed worse than the baseline RotatE model on the FB15k dataset. This suggests that FB15k does not suffer significantly from relation imbalance, making this strategy less beneficial. Given its lack of improvement on our primary dataset, we excluded this module from our final integrated model.

D. Type-Constrained Negative Sampling (Explored, Not Adopted)

We experimented with a **type-constrained negative sampling** strategy, where negative samples that were clearly implausible were discarded. For example, if the corrupted tail entity belonged to a type incompatible with the given relation, that sample was omitted. However, we found that adversarial negative sampling already downweights such trivially false negatives by assigning them low sampling probabilities. This explicit filtering did not improve performance and introduced additional computational overhead. As a result, we excluded this technique from our final model.

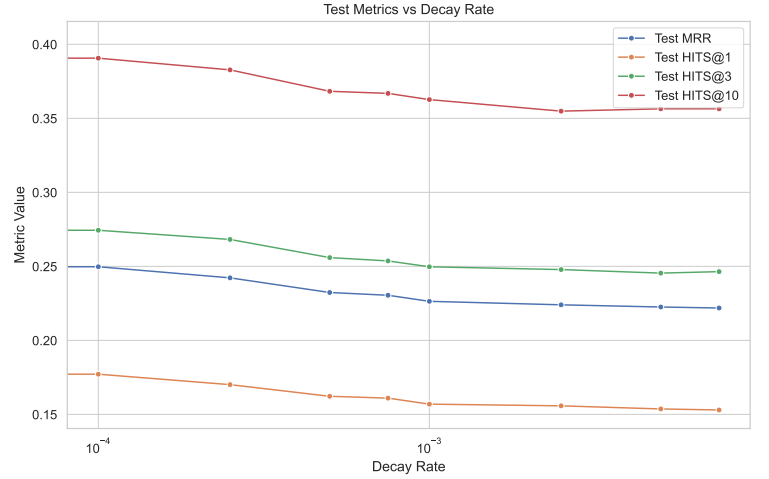


Fig. 2. Visualization of the Adversarial Temperature Annealing

E. Adversarial Temperature Annealing (Our Contribution)

RotatE employs adversarial negative sampling, where the probability of selecting a negative triple is proportional to:

$$P(i) \propto \exp(\alpha \cdot \text{score}_i) \quad (4)$$

The **adversarial temperature** α controls the sharpness of this distribution. A static α can lead to unstable training: low α underemphasizes hard negatives, while high α causes overfitting early on. We introduce a **temperature annealing schedule**:

- Start with a lower α for more diverse negatives during early training.
- Gradually increase α , focusing on harder negatives as the model stabilizes.

This improves convergence and training stability, especially on large-scale graphs.

F. Summary of Modifications

This unified framework enhances RotatE by addressing limitations in both relation representation and training dynamics, laying a foundation for further experimentation on imbalanced and complex knowledge graphs.

IV. CONTRIBUTIONS AND RESULTS

In this study, we aimed to improve the RotatE knowledge graph embedding framework by addressing its limitations in handling non-commutative relations, multi-fold relational patterns, and interpretability. Our primary contributions are as follows:

- 1) **Implementation and Comparison of Advanced Rotational Models:** We systematically implemented and evaluated RotatE, RotatE-CT, and MRotatE within a unified framework. This allowed a precise comparison of their strengths and weaknesses regarding relation representation, interpretability, and performance on multi-fold relational patterns.

TABLE I
ABLATION STUDY RESULTS ON FB15K-237 AT STEP 4999

Model	MRR	MR	HITS@1	HITS@3	HITS@10
RotatE (Base)	0.2175	1010.86	0.1493	0.2395	0.3536
RotatE-CT	0.2748	325.95	0.1899	0.3062	0.4417
MRotatE	0.2964	401.67	0.2100	0.3260	0.4708
RotatE-CT + MRotatE	0.3069	252.93	0.2149	0.3411	0.4892

TABLE II
PERFORMANCE OF RELATION-BALANCED SAMPLING VS. ROTATE (FB15K, STEP 4999)

Model	MRR	MR	HITS@1	HITS@3	HITS@10
RotatE (Base)	0.2175	1010.86	0.1493	0.2395	0.3536
Relation-Balanced Sampling	0.1705	2403.18	0.1352	0.1864	0.2314

TABLE III
SUMMARY OF PROPOSED ENHANCEMENTS TO ROTATE.

Model / Idea	Limitation Addressed	Mechanism
RotatE-CT	Non-commutative relations	Rotation + translation bias
MRotatE	Multi-fold relations	Entity rotation, dual-geometry embeddings
Relation-Balanced Sampling	Relation frequency imbalance	Balanced relation-level triple sampling
Adversarial Temp Annealing	Static negative sampling focus	Dynamic temperature schedule for negatives

- 2) **Entity Rotation Mechanism:** Inspired by MRotatE, we incorporated entity rotation, which constrains the distance between head and tail entities to match the distance of their corresponding relations. This enabled effective modeling of one-to-many, many-to-one, and many-to-many relations, addressing a key deficiency in RotatE and RotatE-CT.
- 3) **Annealed Adversarial Temperature:** Following our analysis of RotatE’s training dynamics, we applied annealing to the adversarial temperature in negative sampling.
- 4) **Comprehensive Evaluation:** Our experimental results demonstrate that the combination of entity rotation and temperature annealing significantly improves performance on benchmark datasets. Specifically, we observed notable gains in Mean Reciprocal Rank (MRR) and Hits@K metrics compared to baseline RotatE model. Additionally, our modifications preserved interpretability by maintaining geometric relations in the embedding space, allowing relational patterns to remain visually and conceptually meaningful.

In summary, our work demonstrates that carefully integrating advanced rotational mechanisms and training improvements can substantially enhance both the predictive performance and interpretability of knowledge graph embeddings. These contributions provide a foundation for future research on expressive yet interpretable KGE models.

V. CONCLUSION AND FUTURE WORKS

In this work, we explored and extended the RotatE family of knowledge graph embedding models, focusing on improving their ability to represent non-commutative and multi-fold

relational patterns while maintaining interpretability. By implementing and evaluating RotatE, RotatE-CT, and MRotatE, we identified key limitations of each model and introduced enhancements such as the entity rotation mechanism and annealed adversarial temperature. Our experimental evaluation demonstrated significant improvements in predictive performance, particularly in Mean Reciprocal Rank (MRR) and Hits@K metrics, while preserving the geometric interpretability that makes rotational embeddings attractive.

Despite these contributions, several limitations remain. First, while entity rotation improves the modeling of many-to-one and one-to-many relations, it introduces additional parameters that may lead to higher computational costs, particularly on large-scale knowledge graphs. Second, the relation-balanced sampling approach, though effective in improving rare relation modeling, requires careful tuning to prevent overfitting to low-frequency relations. Third, our methods focus primarily on geometric interpretability and may not fully address challenges in explainability at a symbolic or reasoning level.

Future research could address these limitations in several directions. One promising path is to explore hybrid models that combine the geometric interpretability of rotational embeddings with symbolic reasoning frameworks, enabling deeper explanations for link predictions. Another direction involves scaling the proposed methods to extremely large graphs, potentially by leveraging distributed training techniques or parameter-efficient embedding strategies. Additionally, integrating other advanced negative sampling schemes or automated hyperparameter optimization could further improve performance while reducing manual effort. Finally, extending this work to temporal and dynamic knowledge graphs represents a compelling opportunity to capture evolving relational

patterns over time.

Overall, this research contributes to the ongoing effort to design expressive, interpretable, and scalable knowledge graph embedding models. Our findings highlight the importance of balancing predictive accuracy with interpretability and provide a strong foundation for future advancements in this field.

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